QQ-A-250/21B
30 March 1984
SUPERSEDING
QQ-A-250/21A
17 June 1974

FEDERAL SPECIFICATION SHEET

ALUMINUM ALLOY, 7178-T76, PLATE AND SHEET (IMPROVED EXFOLIATION RESISTANT)

This specification was approved by the Assistant Administrator, Officer of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

The complete requirements for procuring the aluminum alloy, 7178-T76, plate and sheet for improved exfoliation resistant described herein shall consist of this document and the latest issue of QQ-A-250/GEN (see 2.1).

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specificaion covers the specific requirements for improved exfoliation resistant aluminum alloy, 7178-T76 sheet and 7178-T7651 plate.

1.2 Classification.

1.2.1 Tempers. Sheet shall be furnished in the T76 temper only and plate in the T7651 temper only, as specified (see 6.2). The definition of these tempers shall be in accordance with the general specifciation QQ-A-250/GEN.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specification

QQ-A-250/GEN - Aluminum and Aluminum Alloy Plate and Sheet; General Specification for

Federal Standards

FED-STD-184 - Identification Marking of Aluminum, Magnesium and Titanium

FSC 9535

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards

MIL-STD-2154 - Inspection, Ultrasonic, Wrought Metals, Process for

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards

G34-72 - Exfoliation Corrosion Susceptibility in 7XXX Series Copper Containing Aluminum Alloys (EXCO Test)

G47 - Determining Susceptibility to Stress-Corrosion Cracking of High-Strength Aluminum Alloy Products

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Chemical composition. The chemical composition of the ingot or slabs used for the manufacture of the plates and sheets shall conform to the requirements specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

TABLE I. Chemical composition 1/

Element	Percent	
	Minimum	Maximum
Silicon	-	0.40
Iron	-	0.50
Copper	1.6	2.4
Manganese	-	0.30
Magnesium	2.4	3.1
Chromium	0.18	0.28
Zinc	6.3	7.3
Titanium	-	0.20
Others, each	-	0.05
Others, total 2/	-	0.15
Aluminum	Remainder	

1/ Analysis shall routinely be made only for the elements specifically mentioned in table I. If, however, the presence of other elements is indicated or suspected in amounts greater than the specified limits, further analysis shall be made to determine that these elements are not present in excess of specified limits.
2/ The sum of those "Others" metallic elements, 0.010 percent or more each, expressed to the second decimal before determining the sum.

3.2 Mechanical properties. The mechanical properties perpendicular to the direction of final rolling, except for material less than 9 inches in width, shall conform to the requirements of table II for the temper specified. For material less than 9 inches in width, the mechanical properties parallel to the direction of final rolling shall conform to the requirements of table II for the temper specified.

TABLE II. Mechanical properties

Form and temper	Thickness in inches	Tensile strength, ksi, minimum	Yield strength at 0.2 percent offset, ksi, minimum	Elongation in 2 inches or 4 times D, percent, 1/, 2/ minimum
T76	0.045-0.249	75.0	64.0	8
T7651	0.250-0.499	74.0	63.0	8
	0.500-1.000	73.0	62.0	6

1/ Not required for material 1/2 inch or less in width.
2/ D represents specimen diameter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

3.3 Resistance to exfoliation corrosion and stress-corrosion cracking.

3.3.1 Exfoliation corrosion. Material in the T76 and T7651 tempers, when tested in accordance with 4.2.3, shall show no exfoliation corrosion equivalent to or in excess of that depicted in figure 2 of ASTM G34-72 for category B. The frequency of testing shall be in accordance with the requirements of QQ-A-250/GEN.

3.3.2 Stress-corrosion cracking. Plate, 0.750-inch thickness and thicker, shall exhibit no susceptibility to stress-corrosion cracking when tested in accordance with 4.2.4. The frequency of testing shall be in accordance with the requirements of QQ-A-250/GEN.

3.3.3 Lot acceptance control criteria. Resistance to exfoliation corrosion and stress-corrosion cracking for each lot of material shall be determined by the following criteria (see 4.1.3 and 4.2.5).

(a) Determine electrical conductivity and mechanical properties
(b) If the conductivity is 38.0 percent International Annealed Copper Standards (IACS) or higher, and the tensile properties meet the minimum limits specified herein, the material is acceptable
(c) If the tensile properties meet the minimum limits specified herein, and if the conductivity is at least 35.0 percent IACS but less than 38.0 percent IACS, the material shall be tested as specified in 4.2.3 or reprocessed
(d) If the conductivity is below 35.0 percent IACS, the material is not acceptable and must be reprocessed.

3.4 Internal defects. Unless otherwise specified (see 6.2), plate shall be ultrasonically inspected (see QQ-A-250/GEN). Acceptance limits shall be as specified in table III.

3.5 Marking. In addition to marking as required in FED-STD-184, plate and sheet shall be identified by a lot number in at least one location on each piece.

Table III. Ultrasonic discontinuity acceptance limits 1/

Thickness, inches	Maximum weight per piece (pounds)	Discontinuity class 2/
0.500 through 1.000	2,000	B

1/ Discontinuities in excess of those listed in table III may be allowed subject to the approval of the procuring activity, if it is established that they will be removed by machining or that they are in noncritical areas
2/ The discontinuity class limits are defined in MIL-STD-2154

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS (see QQ-A-250/GEN)

4.1 Sampling.

4.1.1 For chemical analysis. Samples for chemical analysis shall be selected in accordance with QQ-A-250/GEN.

4.1.2 For exfoliation and stress-corrosion cracking tests. Samples for exfoliation and stress-corrosion cracking tests shall be selected in accordance with QQ-A-250/GEN. Thereafter, surveillance testing shall be performed on at least one sample per month and for each size range of sheet and plate produced with acceptable exfoliation resistance. The surveillance test samples shall be taken from a lot which has met the requirements of 3.3.3(b). Samples as above shall be taken for stress-corrosion cracking test when thickness permits (see 3.3.2).

4.1.3 For mechanical property and conductivity tests. Sampling shall be in accordance with QQ-A-250/GEN for mechanical property tests. Mechanical properties and conductivities shall be determined on the same samples (see 4.2.3 and 4.2.5). Conductivity shall be determined prior to testing for mechanical properties.

4.2 Test methods.

4.2.1 Chemical analysis. Samples selected in accordance with 4.1.1 shall be analyzed to determine conformance to 3.1 using the test methods specified in QQ-A-250/GEN.

4.2.2 Mechanical properties. Samples selected in accordance with 4.1.3 shall be prepared for testing and tested in accorance with QQ-A-250/GEN to determine conformance to 3.2.

4.2.3 Exfoliation corrosion test. Samples taken in accordance with 4.1.2 shall be tested for conformance to 3.3.1 using the test method specified by ASTM G34-72. Full thickness specimens of sheet and plate large enough to provide visual assessment of attack may be used; however, for all materials, 0.100 inch or more in thickness, 10 percent of the thickness shall be removed by machining one surface. The machined surface shall be evaluated.

4.2.4 Stress-corrosion cracking tests. In order to determine conformance to 3.3.2, stress-corrosion cracking tests shall be performed in accordance with ASTM G47. The specimens shall be stressed in the short transverse direction with respect to grain flow and held at constant strain. The stress level shall be 25 ksi and test duration shall be 20 days.

4.2.5 Electrical conductivity tests. The conductivity shall be determined by taking three electrical conductivity readings at random on each sample. For thicknesses less than 0.100 inch, the conductivity shall be determined on the surface. For thicknesses of 0.100 inch and over, the conductivity shall be determined on the machined surface after machining one surface of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

sample to a depth of approximately 10 percent of the product thickness. The surface machining may be accomplished by chemical milling. The average of the three readings shall be used as the conductivity acceptance criteria of 3.3.3.

4.3 Records. The producer shall maintain records of the performance of all lots sampled and tested. Upon request of the procuring activity, such records shall be made available.

5. PREPARATION FOR DELIVERY (see QQ-A-250/GEN)

6. NOTES

6.1 Intended use. This alloy is intended for use where high strength is required and where exfoliation and stress-corrosion resistance superior to 7178-T6 is required in high performance military aircraft.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification
(b) Form and temper required (see 1.2)
(c) Dimensions required
(d) Requirements for sizes not specifically covered (see QQ-A-250/GEN)
(e) Special end use requirements
(f) Whether ultrasonic inspection of plate is waived (see 3.4)
(g) Selection of applicable levels of preservation, packaging, and packing required (see QQ-A-250/GEN)

MILITARY INTEREST:

Custodians:

Army - MR
Navy - AS
Air Force - 20

Review activities:

Army - ME
Air Force - 99
DLA - IS

User activities:

Navy - OS, SH

CIVIL AGENCY COORDINATING ACTIVITYES:

GSA-FSS
NASA-JFK, MSF
DOE-BPA

PREPARING ACTIVITY:

NAVY - AS

DoD Project No. 9535-0350

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.